BOARD INSPECTION UPDATE

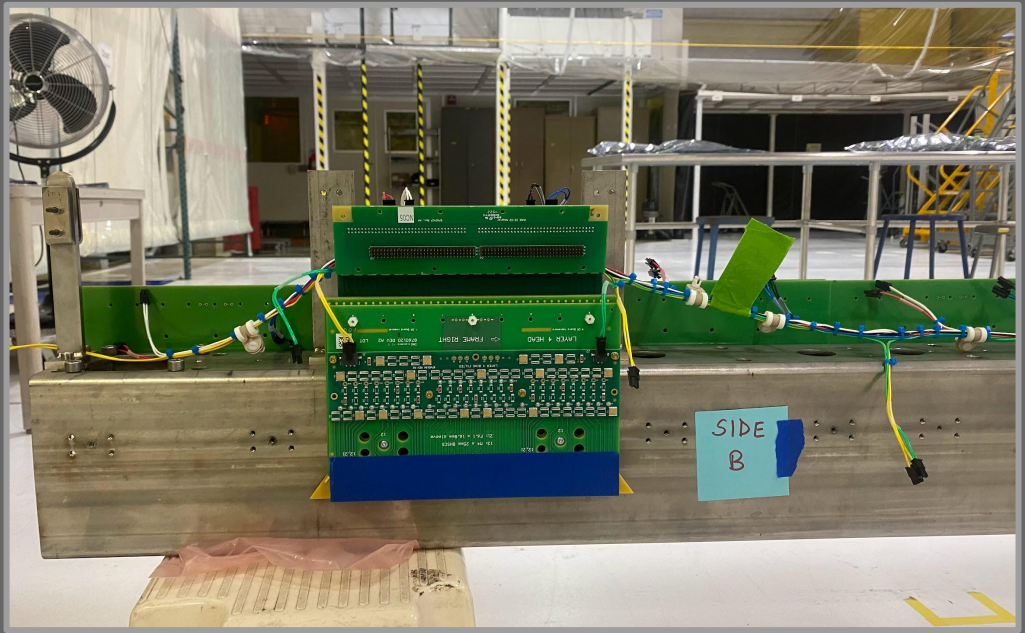
•••

Krishna Lakkaraju
9/11/24

Physical Sciences Laboratory
University of Wisconsin-Madison

OVERVIEW

- UW Populates: GBias, CR, CE-Adapter, and SHV
- Inspect Blank Boards
  - Looking for visual deformities like scratches
  - Measure distances between important components



*Model Head Tube Assembly with Sample Board Stack

NOTE: All Project Progress Made was in Conjunction with, and Made Possible by, the Staff at UW and PSL

INSPECTION QUANTITIES

*Incoming Board Inspection Storage/Staging Shelf

	QTY	# INSPECTED	MANUFACTURER
GBIAS	1000	1000	EPEC
CR	1000	~10	IMAGINEERING
ADAPTER	1000	164	EPEC

MEASUREMENT DEVICE

*Photo of the Keyence Measurement Device

Measurement date/time	8/27/2024 12:14:52 PM
Lot number	G Bias
Serial counter	0834
Name	PSL Incoming Inspection
Product name	8760196 Rev. A2 G Bias
Storage destination	
Figure number	Date Code 2324
Process	Pack Date 24-07-18
Measurement device	VR-6200#AC0N000003
Overall result	OK
Remarks	PO MSN0565368 EPEC 7Aug2024

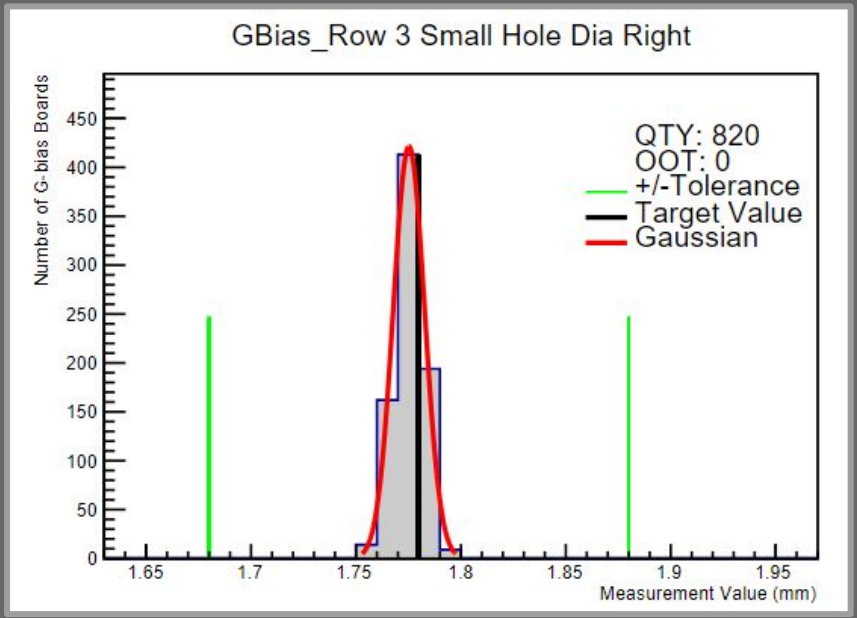
Page 1

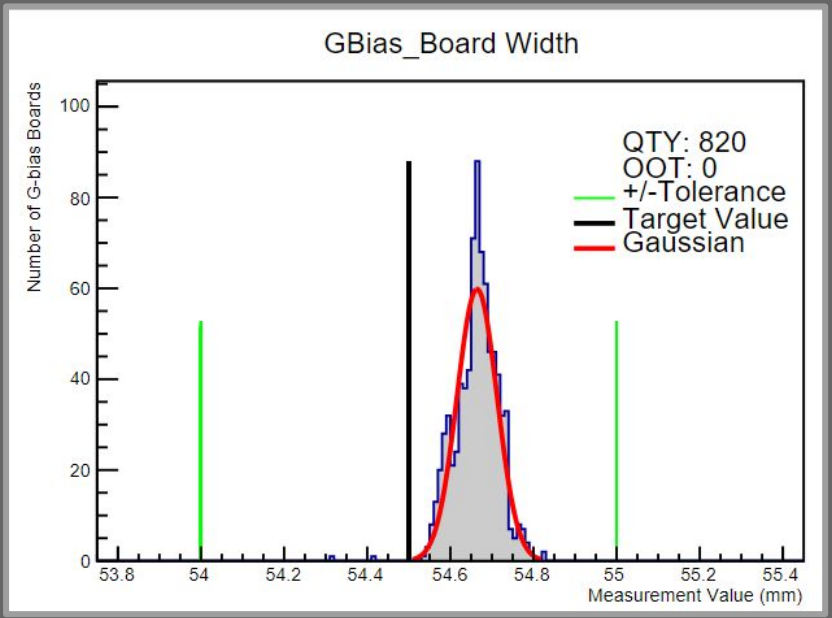
<Measurement result>

No.	Measurement name	Element 1		Element 2	Comment	Judgement	Measurement result	Unit	Design value
1	Plane(Point height)		1	Board Length		OK	227.914	mm	228.00
2	Plane(Point height)		2	Board Width		OK	54.603	mm	54.50
3	Plane(Point height)		3	Hole 1 Diameter		OK	3.253	mm	3.20
4	Plane(Point height)		4	Hole 2 Diameter		OK	3.255	mm	3.20
5	Plane(Point height)		5	Hole 3 Diameter		OK	3.254	mm	3.20
6	Plane(Point height)		6	Hole 4 Diameter		OK	3.251	mm	3.20
7	Plane(Point height)		7	Hole 5 Diameter		OK	3.258	mm	3.20
8	Plane(Point height)		8	Hole 6 Diameter		OK	3.256	mm	3.20
9	Plane(Point height)		9	Hole 7 Diameter		OK	3.253	mm	3.20
10	Plane(Point height)		18	Hole 5 X		OK	109.517	mm	109.50
11	Plane(Point height)		19	Hole 7 X		OK	109.505	mm	109.50
12	Plane(Point height)		11	Hole 1 Y		OK	29.011	mm	29.00
13	Plane(Point height)		17	Hole 4 Y		OK	29.002	mm	29.00
14	Plane(Point height)		20	Row 1 Small Hole Dia Left		OK	1.760	mm	1.78
15	Plane(Point height)		21	Row 1 Small Hole Dia Rt		OK	1.763	mm	1.78
16	Plane(Point height)		22	Row 2 Small Hole Dia Left		OK	1.767	mm	1.78
17	Plane(Point height)		23	Row 2 Small Hole Dia Rt		OK	1.772	mm	1.78
18	Plane(Point height)		24	Row 3 Small Hole Dia Left		OK	1.768	mm	1.78
19	Plane(Point height)		25	Row 3 Small Hole Dia Right		OK	1.761	mm	1.78
20	Plane(Point height)		26	Row 1 Small Hole Y		OK	38.506	mm	38.50

*Sample of a Keyence Measurement Output File

MEASUREMENT PLOTS

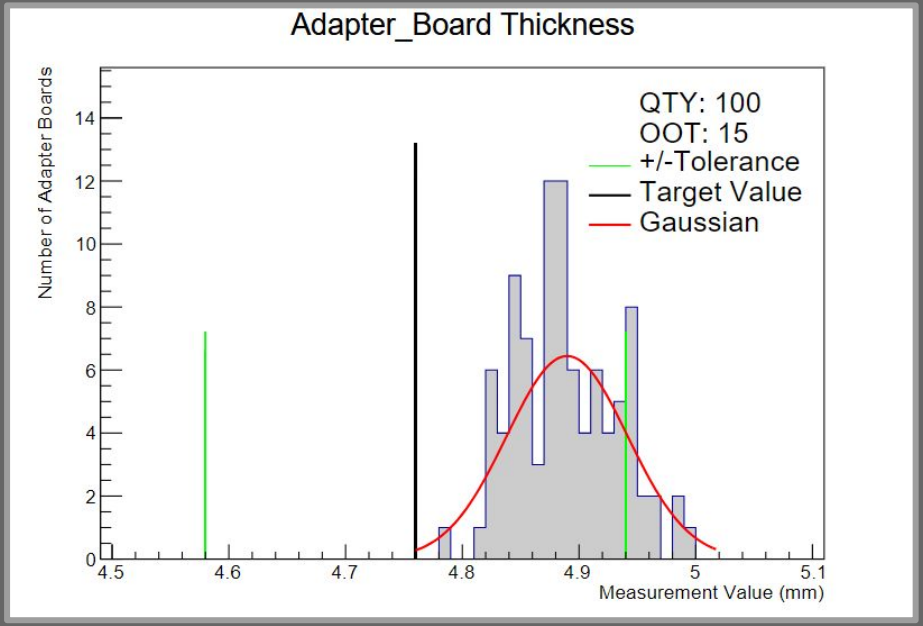




*Example Plots Illustrating Acceptable Distributions

- Green Bars: +/- Tolerance Values
- Black Line: Nominal Value
- Red Graph: Fitted Gaussian Distribution
- OOT: “Out-Of-Tolerance”

CE-ADAPTER THICKNESS



*Histogram Showing Out-Of-Tolerance (OOT) Measurement Values

- 15% Of Adapter Boards Inspected (100) in the Latest Shipment are too Thick
- Board Thickness May Influence Connector Clearance
  - Impacts Electrical/Physical Connection
- All Boards Were Accepted
  - Electrical/Physical Tests have been Conducted using IPC-6012 Guidelines
  - Four Boards have been Assembled

CURRENT PROJECTS

...

CE-ADAPTER TEST BENCH

- Used After Component Population
- Utilizes Two Arduinos and a Series of Multiplexers
- Tests Isolation/Mapping
  - In Addition, it will Determine Effects of Thick Boards

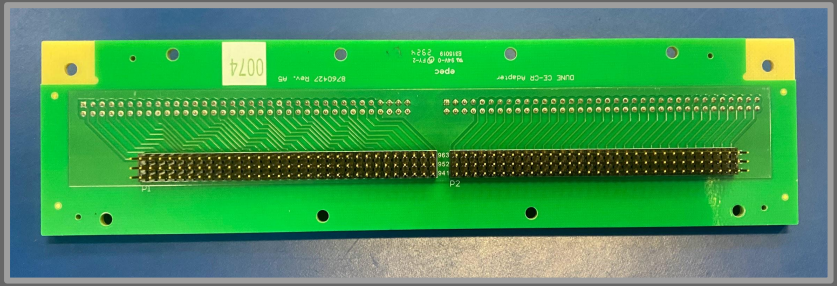


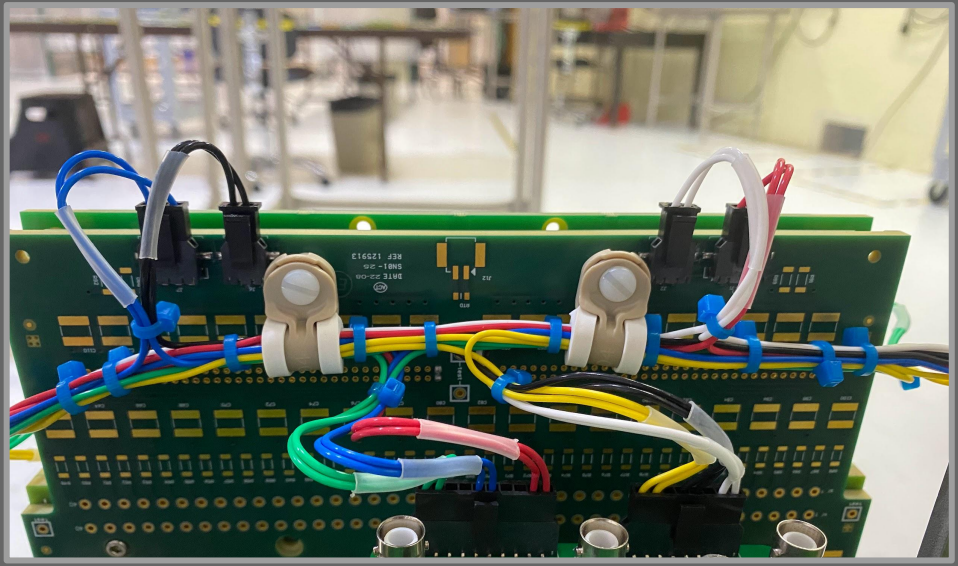
*Photo of A CE-Adapter Board and its Connectors

```
Testing 96-pin Connector for Isolation...
Error: Signal detected on Pin 33 indicating a failure in isolation.
Error: Signal detected on Pin 34 indicating a failure in isolation.
Error: Signal detected on Pin 35 indicating a failure in isolation.
Error: Signal detected on Pin 36 indicating a failure in isolation.
Error: Signal detected on Pin 37 indicating a failure in isolation.
Error: Signal detected on Pin 38 indicating a failure in isolation.
Error: Signal detected on Pin 39 indicating a failure in isolation.
```

***Mock** Serial Output from the Test Bench Code

HARNESS PRODUCTION

- Distributes Voltage to Layers
- 14 Sides/7 Sets Fabricated
  - Pending Testing
- Developing Procedures/Test Systems
  - New Automated Crimper/Strippers



*Interior View of the Boardstack with an Attached Wire Harness

CONCLUSION

- Now able to Easily Analyze Incoming Board Inspection Data
- Some CE-Adapter Boards are Beyond Specification Tolerance
  - We have Accepted Them for Production
- An Adapter Board Test Stand is Being Developed to Acquire More QA Data
- Wire Harness Production has Begun